

Introduction to DDoS

Module 46

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dear viewers today we are going to discuss one serious kind of Cyber attack known as distributed denial of service attack we will try to explore about introduction of Dos attacks motivation behind these attacks what are the consequences of these attacks vulnerabilities in the basic protocols that can be exploited by the attackers to launch these attacks and in the last part of the lecture we will discuss about the classifications of these attacks there has been a tremendous growth of Internet especially in the last decade in terms of its usage that has led Our Lives to become increasingly dependent on the internet the internet has become essential for information flow business Commerce communication

education entertainment and research activities initially the major goal of the internet development was to enhance the unobstructed free flow of data and scalability without keeping mind the security issues which now appears to be precious its architectural design Was Made Simple for ease of usage but subsequently it fascinated the attackers who exploit the vulnerabilities and launch a variety of attacks for plethora of reasons these threats can Target basic CIA model of security which deal with confidentiality of data Integrity of data and availability of services availability of services is primarily targeted by launching a denial of service attacks which is characterized by an explicit attempt by attackers to

prevent legitimate users of service from using that service denial of service attacks has undoubtedly take over taken all concerns and has emerged as arguably the most severe form of attacks in the last decade the major goal of Dos attacks is to disrupt or deny the services for numerous reasons think out a situation you are a reputed online shopping company and suddenly your web service becomes unavailable to the customers when you deeply investigate you realizes that a flood of packets is surging into your network you have just become one of the hundreds or thousands of victims of denial of service attacks a pervasive and growing threat to the internet traditionally the security Community has focused its attention on unauthorized disclosure or modification of information and perhaps theft of

services denial of service was largely ignored as being unlikely to occur because the characters would not gain anything from such an attack clearly this is not the case today denial of services can be devast deating attack it 3 minutes, 1 second can put Merchants out of

business and can cause major and very visible disruption to our cyber world it can be used against specific companies for which the attacker has a grudge or has been paid to attack or it can be used by terrorists to cause major disruption to critical intermediate and victim infrastructure this type of attack may include attempts to flood a network disrupt connections between two machines disrupt service to a specific system or person thereby preventing legitimate Network traffic to access Network server

service or website thus a denial of service attack aims to deny access by legitimate users to services or resources such an attack can be configured in many ways like targeting operating system or Network Services the first way targets to crash a system by sending one or more carefully crafted packets that exploit 4 minutes, 1 second software vulnerabilities On Target server for example the Ping of death attack that sends large International control message protocol ping packets that is fragmented into multiple datagrams to Target system which can cause certain operating systems to crash freeze or reboot due to buffer overflow the second way is known as flooding type of Dos attacks that employs massive volumes of useless traffic to occupy all the resources that

could hamper service to legitimate traffic while it is possible to prevent the first form of attack by patching non vulnerabilities however the second form of attack cannot be easily prevented the targets can be attacked simply because they are connected to the public internet next comes the bigger version of denial of service attacks called distributed denial of service attack when the traffic of Doss attacks comes from multiple sources it is called distributed denial of service attack by using multiple sources the power of a Dos attack is Amplified and the problem of defense is made much more complicated both dos and dos are a huge threat to the operation of internet sites but the DS problem is more complex

and harder to solve because of the following reasons first it uses a very large number of machines this yields a powerful weapon any Target regardless of how well provisioned it is can be taken offline Gathering and engaging a large army of machines has become terribly simple because many automated tools for DS attacks can be found on hacker web pages and in chat rooms such tools do not require sophistications to be used and can inflict very effective damage a large number of machines gives another advantage to an attacker even if the target were able to identify attacking machines and there are effective ways of fighting this information what action can be taken against a network of 1,000 host the second characteristics of some of the Dos attacks that increase their complexity is the use of seemingly

legitimate traffic resources are consumed by large number of legitimate looking messages when comparing the attack messages with legitimate one there are frequently no feature to distinguish them since the attack misuses a legitimate activity it is extremely hard to respond to the attack without also disturbing this legitimate activity look at the figure which shows the complete mimicry of the DS attack the operating system and the network Protocols are

developed without applying security engineering which results in providing hackers a lot of insecure machines on the internet these insecure unpatched machines are used by DoS attackers as the Army to launch attack an attacker or a hacker gradually implants attack programs on these

insecure machines depending upon sophistications and logic of implanted programs these compromised machines are called Master's handlers or zombies and collectively called Bots and attack network is called botnet in the hackers Community hackers send control instructions to the Masters which in turn communicate it to zombies for launching attacks the zombie machines under the control of Masters and handlers as shown in the figure transmit attack packets which converge at victim or its Network to exhaust either its communication or computational resources numerous incidents of DoS attacks are common these days from a normal client users to business communities ISPs and also the world

famous companies such as Yahoo Twitter Facebook eBay Google are the victim of DoS attacks a few recent and famous DoS incidents are shown in the table the table describes about the incidents in chronological order right from 2000 to 2014 along with the description of attacks it is clear from the table that DoS attacks is a sensitive issue prevailing among the cyber community and also it has been analyzed from the recent security reports that these attacks have gained a lot of strength in recent years that is from 2014 onwards as compared to earlier before year 2005 the attack size was less than 10 Gbps and in 2010 it reached a mark of 100 Gbps furthermore the attack size was observed up to 400 Gbps in 2014 which shows the severity of problem over the internet now the question comes in our

mind that why a hacker launch such kinds of attacks what are their motives so in this coming discussion we will be exploring some of the key factors that contribute towards launching of these hostile activities on the internet first point to be discussed about the recognition some of the early DoS attacks were largely pranks or simple pranks played by hackers the ultimate goal was to prove that something could be done such as taking a popular website offline such a major achievement brings an attacker recognition in the underground cyber community next is ideological activism another frequent motive of DoS attacks is self-described as being political individuals or groups who disagree with views or actions of certain organization

an online media site a corporation or a government have been known to launch DoS attacks against computers and networks owned by this organization financial gain can be another factor which motivates the attackers through the monetary gain to launch several kinds of DoS attacks after launching these attacks they may also demand for a ransom amount of money to prevent further attacks the attackers involved these types of are most experienced next is vandalism the motive of this category of attacker is to destroy the available resources as they don't believe in anything and are senseless according to the recent reports the attackers related to this type of attacks have been increased recently lastly revenge or competitive spirit

in the business attackers may

launch the Dos attacks to take the Revenge the attackers may be employees of certain organizations that were Dishonored in some dispute however there is a major lack of data on preparators and motives of Dos attacks the vast majority of attacks are not reported in further discussion we will be exploring what are the consequences of these attacks the consequences of do attacks may be very severe it depends on the kind of attack launched by the attackers the organization and the target of the attackers the long duration dos attacks are more rigorous and these must be mitigated quickly after detection DS attacks may lead to the following consequences huge revenue losses reputation of organization may go down loss of shareholders confidence customers dissatisfaction loss of important data

due to system crashes extra operational cost hence consequences of these attacks are so severe that it may lead to social financial and legal implication among the cyber community let's now discussed about the vulnerabilities present in our basic protocols and architecture that lead to perform this type of sophisticated attack firstly the stateless nature of the internet it means that intermediate routers do not maintain any straight information about the forwarded packets because of this accountability computer forensics are difficult to conduct that theft that Ur unidentified secondly there is a lack of authenticity over the internet without authentication malicious internet users can claim the identities of other users

without being easily detected or located the most common example of this type of attack is IP spoofing where attacker can easily spoof The Source IP address of the another machine thirdly the deterministic nature of the Internet Protocol this is not a design flaw but is often necessary to proper operation of Internet protocols that is the TCP connection establishment procedure and TCP connection control mechanism the three-way handshake TCP mode mechanism which is predictable and can be easily exploited to launch flooding type of attack known as Sin flooding attack last but not the least is the mimicking legitimate user Behavior attackers can easily mimic the behavior of legitimate users which leads

them to perform a kind of attack called application layer dos attacks in the last last part of the lecture we will be discussing about the classification of D attacks D do attacks may occur at every layer of OSI model therefore these can be classified as follows as shown in the figure physical layer attack volumetric attacks application layer attacks so first comes the physical layer attacks also known as layer one attack these types of attacks are concerned with physical security these attacks includes the destruction obstruction malfunctioning or manipulation of physical assets at the physical level the communication relies on nodes host switches hubs routers and server and the medium of transmission

that can be wired or Wireless if the devices through which the services are granted gets down there will be a disruption of services this target can be easily achieved if the signals are interrupted between communicating devices to prevent them from reaching to their

destinations so the attackers may attain their targets of denying the services by compromising the physical security of the assets the various kinds of physical attacks destruction of Assets in this kind of attack the attacker may damage the physical assets or may cut down the backbone wires of local area network so that Services can be denied by this the availability of these assets could be reduced and there may become unresponsive resulting in denying of services to prevent such kind of an attack physical security of assets is

required next is jamming the jamming attack is executed primarily in the wireless sensory networks jammers can be used by attackers to hamper the functioning of the wireless networks by generating noise signals close to the network by doing this the throughput of the network may fall drastically in Jamming attacks packets may be corrupted by generating interference between the communicating nodes next is tempering the attackers may temper with the functioning of nodes by erasing all programs or by modifying the codes for assessing the information of high layers cryptographic keys and other sensitive information may be extracted by compromising the nodes and keeping the packages temper approved the tempering attack can be prevented next category is volumetric

attack also known as layer two to four attacks in this attack congestion is caused by consuming the bandwidth of the network through flooding attacks like mac flooding ICMP flooding UDP flooding SYN flooding Etc therefore these attacks cause high traffic to the network a large number of DDoS attacks variants fall under this category of volumetric attacks the basic protocol of this attack category is to congest the network or Target machine with large amount of packets that is flooding the target this results in total consumption of resources by the attacker moreover the targeted machine may become unresponsive and possibly it may crash completely the attacker transmits such kind of amount of data toward the target Target machine through well distributed

and Scattered Bots that are remotely controlled by the bot Masters the key features of these attacks is the most of the IP packets used have spoofed Source IP addresses the spoofing of addresses in the IP packets protect the actual attackers from being traced back the different kinds of volumetric attacks are Mac flooding icmp flooding UDP flooding the attackers clone the Mac addresses and send a huge number of frames of data towards the switches besides the non-existing Mac addresses are used by The Tackle to alter the ARP caches switches have a certain memory that is used to handle the request and to reserve the resources for them when huge volume of forged requests are received by the switches it starts allocating the resources for the cloned

Mac addresses and start denying the request to the legit MAC ones likewise it may also affect the routing tables of the routers at the network to drop the entries in the table which make the routing Services unavailable next comes the icmp flooding icmp flood is the error control protocol in the network layer query messages are used to diagnose the network problem through icmp requests that are encapsulated into IP packet attackers send numerous ICMP MP

packets to the Target server with huge range of spoofed source IP addresses the target servers in turn replies to these queries which results in consumption of bandwidth on the network consequently the network gets congested and becomes unresponsive. Additionally the fragmented icmp packets are also used by the attackers with very high sending rate

such that server gets overloaded by This Server key keeps resembling to fake data packets and legitimate users do not get the services. Smurf attacks and the Ping of Death attacks are the common examples which make use of icmp protocol. Next is the TCP flooding. The attackers exploit the vulnerability of three-way handshake mechanism in the TCP protocol of the transport layer to launch such kinds of attack. In a flooding attack, fragmented acknowledgement packets are examples of TCP flood attack. In a sin flooding attack, the attackers send large number of send requests to the Target server without spoof address to establish the connection. Upon receiving the S request, the server allocates the resources to each request and sends response with S A

packets as responses of the server will not be entertained due to the spoofed addresses and TCP 3way handshake connections remain in half open State till time out. A very large number of such requests are sent by the attackers to make the target machine unavailable. The scenario of sin while flooding attack is shown in the figure. Next is the UDP flood. Unlike TCP, UDP is connectionless. Protocols of Transport layer UDP packets are used by attackers to exhaust the bandwidth of network by launching the attack from distributed side with the help of bots. Attack converges high traffic towards the victim. The Echo and the charging Services of UDP are generally exploited to launch such kind of attack. Other kinds of attacks which can be launched through UDP are DNS flood

UDP fragmentation, viip flood, Etc. Final classification is based on the application layer attacks which are also known as layer five to layer seven attacks. In contrast to volumetric attacks, these attacks generate low traffic at the network but send the large request to the web server. Hence these attacks can critically overwhelm the resources of the web server. These attacks include HTTP get and post request flooding. Etc. Various Protocols of the application like HTTP, https and DNS, Etc. are being exploited by the attacker. At Last I would like to conclude that 21 minutes, 1 second dos attacks is a devastating attack among the cyber community though there are many commercial Solutions present in the market but the ideal solution to this problem is eluded so far. In the next lecture we will discuss

about the different approaches which by which we can tackle about the dos problem. Thank you